

Keyframing

Keyframing is the basis to animate an object. Using Keyframe, motion can be added to the static object because any object has an initial state that changes over time as regards the color, form, position, luminosity, and other properties to form the final state of the object. The basic idea behind keyframing is that you have to add the conditions or the keyframes describing the object transformation. The intermediate positions of an object will be created using these conditions. Consider an example of the movement of a box as shown in the images below:



A) Beginning of Box B) End of Box C) Tweening

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In the first image, the initial position of the box is given, and the second image shows the final position of the box. The total distance between an initial position and the final position is considered to find the intermediate positions. The distance gets divided by the number of intermediate frames, which is five in our case. Hence, the intermediate frames get placed in between at an equal distance calculated by dividing the total distance by the total number of frames. The process to find the intermediate positions of frames to place in between is called keyframe interpolation, tweening, or in-betweening. When all the frames play in succession, the keyframed animation gets generated.

Types of Keyframe Interpolations

Interpolation refers to the method of filling the space between two end-point values. The intermediate movement occurs as you want it to be. Mentioned below are the types of interpolations that can occur between two objects:

- **Linear Interpolation:** This type creates robotic change from the start to end position.
- **Hold Interpolation:** In this type, the sudden changes in the keyframes can be noticed.
- **Ease or Bezier Interpolation:** This interpolation occurs using the movement curve that is adjusted by the user manually.